CHAPTER 05

Alcohols

Independent exercise solving

Original app content and actual Chromium screenshots. No textbook passages or textbook figures are embedded. Source labels and curriculum headings are not evidence of factual correctness.

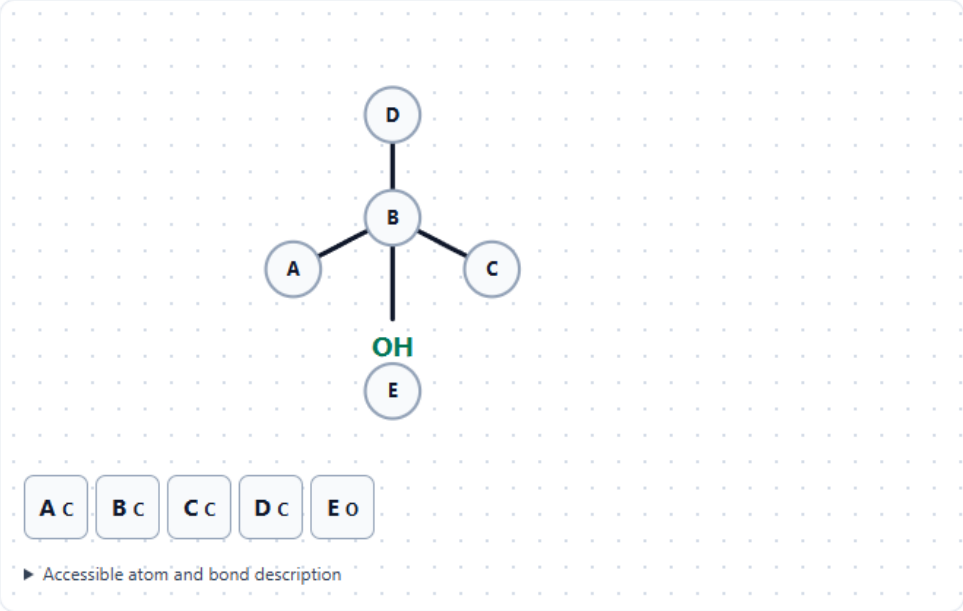
Source snapshot SHA-256: 7c5b4be69e98e33c9a2b0e4cb4a05c20b36132c8aba866a65b6559f1080f8621

BLIND PASS: answer keys, author molecule names/formulas, hints, explanations and answer overlays are withheld. Solve every numbered stage independently. Report the stage ID, answer, derivation, and uncertainty. Question wording and displayed diagram captions are retained because they are visible task data.

Do not open the full review packet in this chat before saving your independent attempt. The full packet contains keys and lessons that would contaminate a blind attempt.

classify-alcohol | Independent exercise

Screenshot: classify-alcohol-stage-1-molecule



Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=166.00, y=115.00; charge=0; explicit H override=none

3 (D): C; x=108.00, y=25.00; charge=0; explicit H override=none

4 (E): O; x=108.00, y=160.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [1, 3, 1], [1, 4, 1]]

classify-alcohol:stage-1 | Select the OH-bearing carbon.

Answer type: atoms

classify-alcohol:stage-2 | How many carbon neighbors does it have?

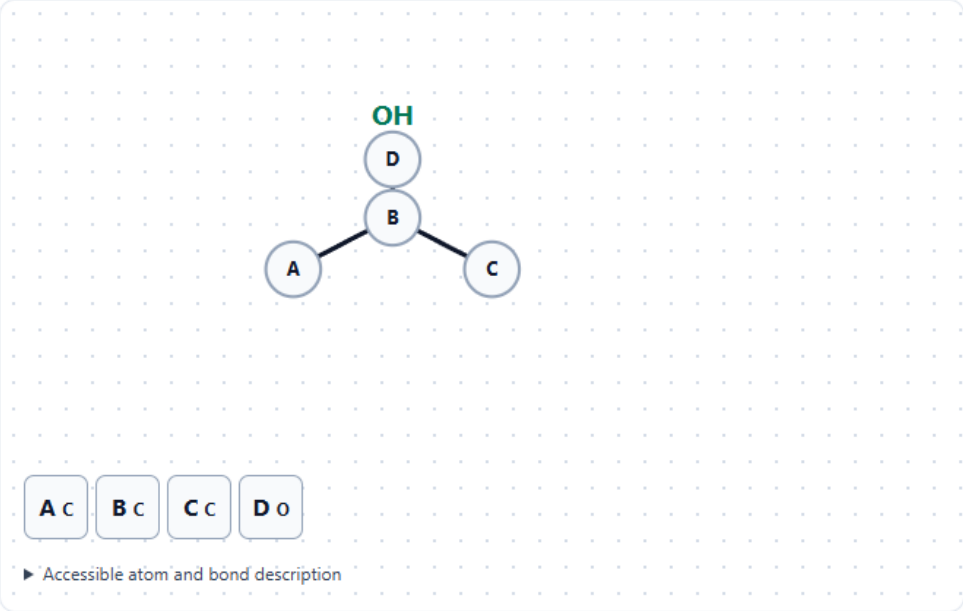
Answer type: number | unit: carbon neighbors

classify-alcohol:stage-3 | Classify this alcohol: primary, secondary, or tertiary.

Answer type: text

oxidize-secondary | Independent exercise

Screenshot: oxidize-secondary-stage-1-molecule



Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=166.00, y=115.00; charge=0; explicit H override=none

3 (D): O; x=108.00, y=25.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [1, 3, 1]]

oxidize-secondary:stage-1 | Select the carbon that changes oxidation state.

Answer type: atoms

oxidize-secondary:stage-2 | Ordinary oxidation produces which functional group?

Answer type: choice

Option 0 (A): Ketone

Option 1 (B): Aldehyde

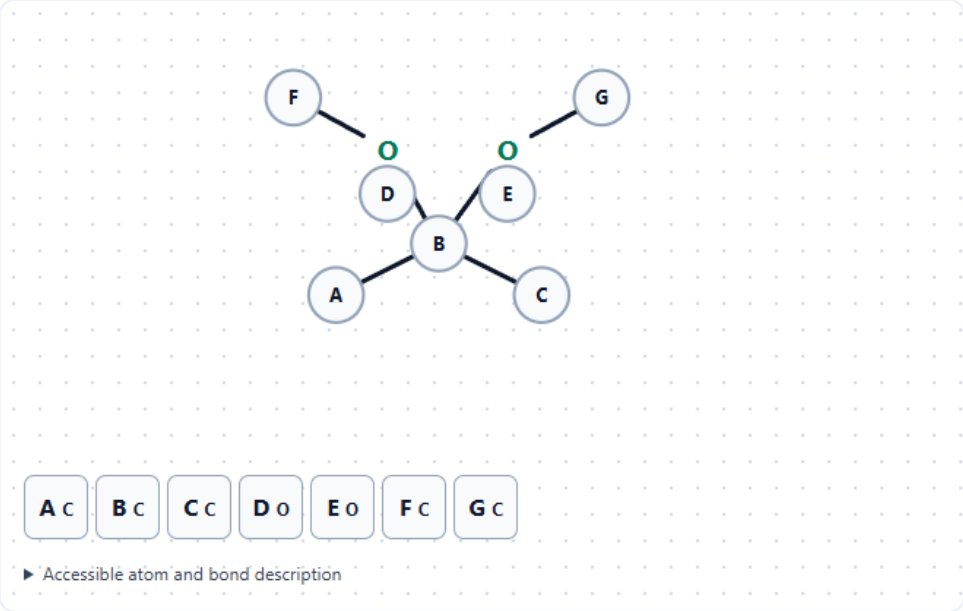
Option 2 (C): Carboxylic acid without C–C cleavage

oxidize-secondary:stage-3 | Name the three-carbon product.

Answer type: text

protection | Independent exercise

Screenshot: protection-stage-1-molecule



Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=75.00, y=130.00; charge=0; explicit H override=none

1 (B): C; x=135.00, y=100.00; charge=0; explicit H override=none

2 (C): C; x=195.00, y=130.00; charge=0; explicit H override=none

3 (D): O; x=105.00, y=45.00; charge=0; explicit H override=none

4 (E): O; x=175.00, y=45.00; charge=0; explicit H override=none

5 (F): C; x=50.00, y=15.00; charge=0; explicit H override=none

6 (G): C; x=230.00, y=15.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [1, 3, 1], [1, 4, 1], [3, 5, 1], [4, 6, 1]]

protection:stage-1 | Select the carbon bearing both OR groups.

Answer type: atoms

protection:stage-2 | Which conditions ordinarily regenerate the ketone?

Answer type: choice

Option 0 (A): Aqueous acid

Option 1 (B): Dry neutral solvent alone

Option 2 (C): Base alone under ordinary conditions

protection:stage-3 | Making an alkyl tosylate from ROH directly causes…

Answer type: choice

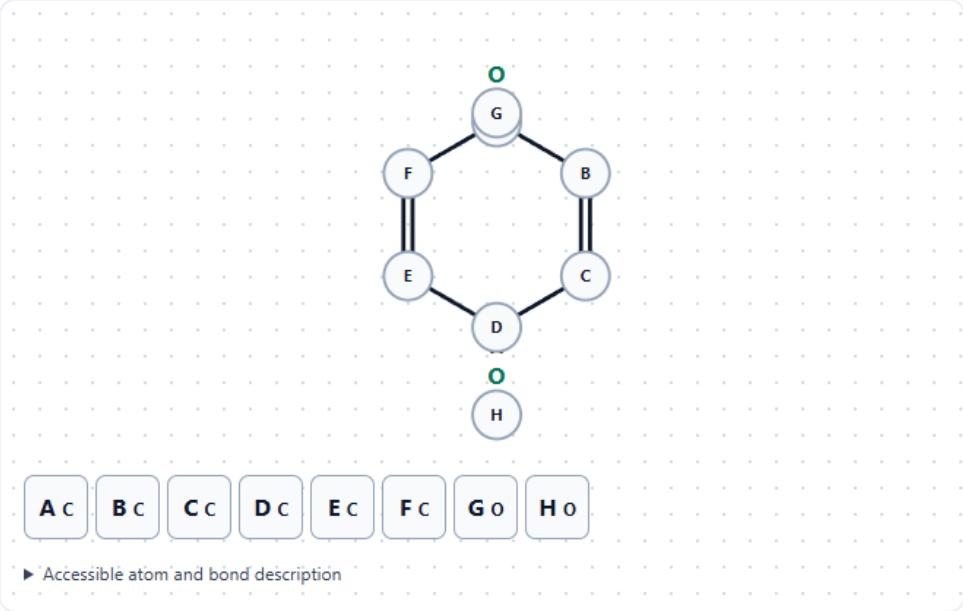
Option 0 (A): Retention of the C–O framework

Option 1 (B): SN2 inversion at carbon in every case

Option 2 (C): Cleavage of the carbon skeleton

quinone-redox | Independent exercise

Screenshot: quinone-redox-stage-1-molecule



Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=170.00, y=37.00; charge=0; explicit H override=none

1 (B): C; x=228.89, y=71.00; charge=0; explicit H override=none

2 (C): C; x=228.89, y=139.00; charge=0; explicit H override=none

3 (D): C; x=170.00, y=173.00; charge=0; explicit H override=none

4 (E): C; x=111.11, y=139.00; charge=0; explicit H override=none

5 (F): C; x=111.11, y=71.00; charge=0; explicit H override=none

6 (G): O; x=170.00, y=5.00; charge=0; explicit H override=none

7 (H): O; x=170.00, y=205.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 2], [2, 3, 1], [3, 4, 1], [4, 5, 2], [5, 0, 1], [0, 6, 2], [3, 7, 2]]

quinone-redox:stage-1 | Select both carbonyl oxygen atoms.

Answer type: atoms

quinone-redox:stage-2 | Quinone → hydroquinone is…

Answer type: choice

Option 0 (A): Reduction

Option 1 (B): Oxidation

Option 2 (C): Only a conformational change

quinone-redox:stage-3 | How many electrons are accepted in the full quinone/hydroquinone conversion?

Answer type: number | unit: electrons